Edexcel IGCSE Physics

Your notes

Components in Series & Parallel Circuits

Contents

Your notes

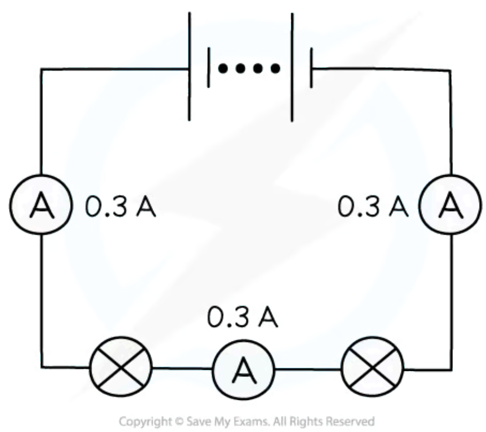
Current in Series & Parallel

Current in series circuits

- There are two ways of joining electrical components:
  - in **series**
  - in **parallel**

Current in series

- A **series** circuit is a circuit that has only one loop, or one path that the electrons can take
- In a series circuit, the current has the **same value at any point**
  - This is because the electrons have only one path they can take
  - Therefore, the number of electrons passing a fixed point per unit time is the same at all locations
- This means that **all** components in a series circuit have the same current

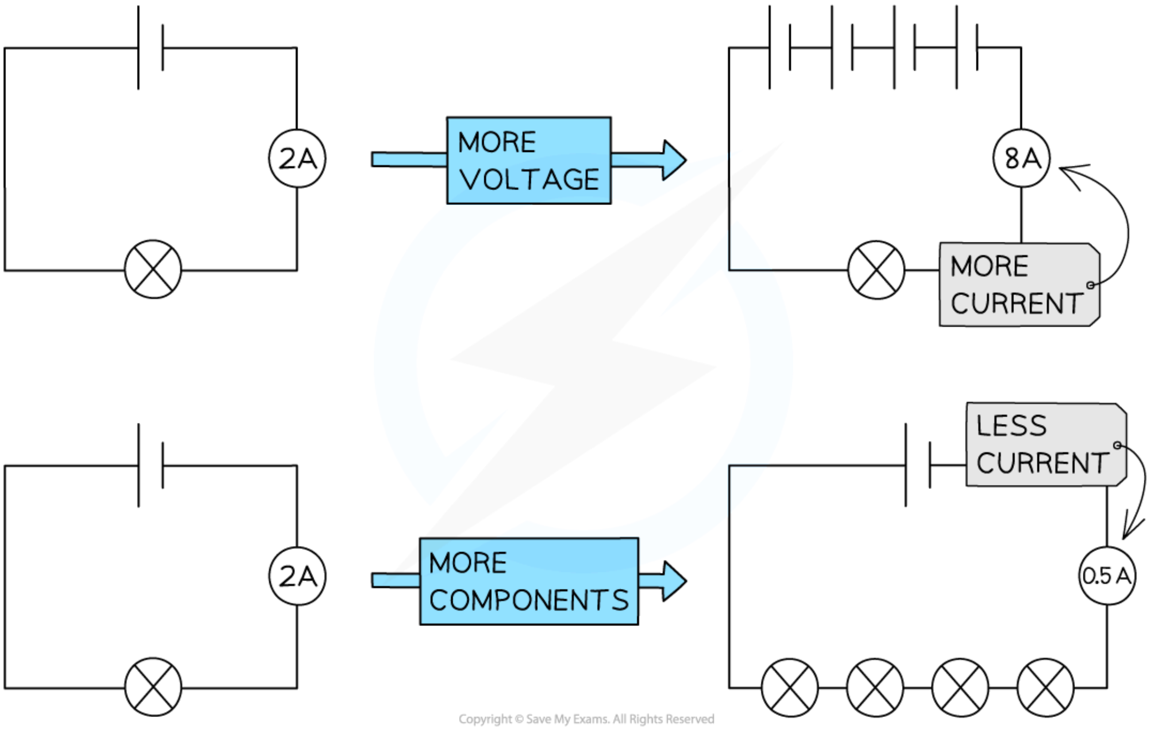


The current is the same at each point in a series circuit

- The amount of current flowing in a series circuit depends on:
  - the **voltage** of the power source
  - the **number** (and type) of components
- **Increasing** the **voltage** of the power source drives **more current** around the circuit
  - So, decreasing the voltage of the power source reduces the current
- **Increasing** the **number** of components in the circuit **increases** the **total resistance**
  - Hence **less current** flows through the circuit

Increasing the voltage and number of components in series

Your notes



Current will increase if the voltage of the power supply increases and decreases if the number of components increases

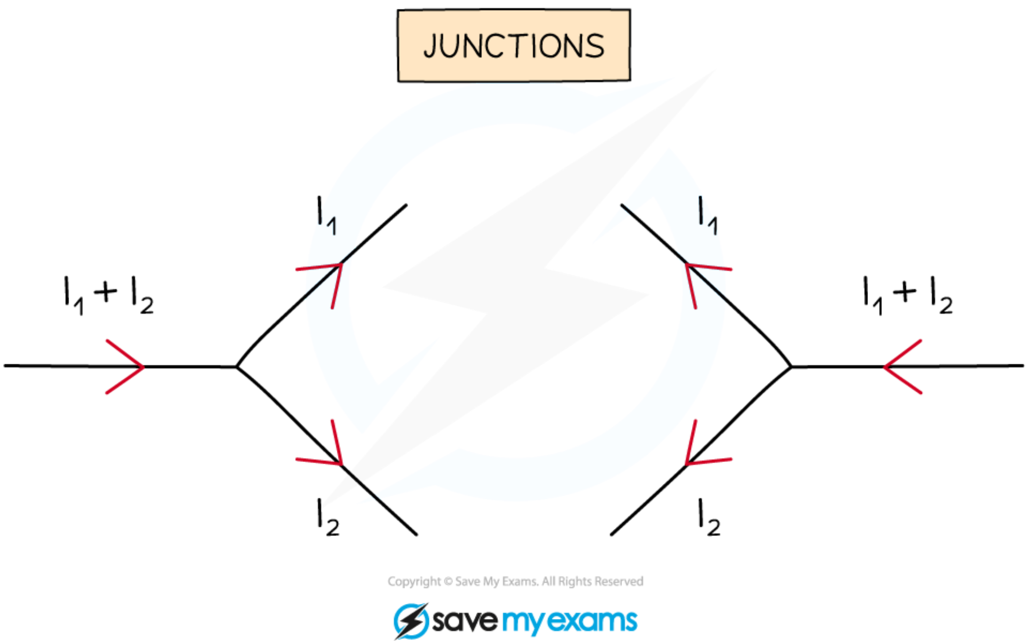
Your notes

Current in parallel circuits

- A parallel circuit is a circuit that has two or more loops, or more than one path that electrons can take
- Parallel circuits contain **junctions** and **branches**
  - Junctions are points where two or more wires meet to form a new branch
  - Branches are the sections of wire between junctions

Current in parallel

- In a parallel circuit, the current has different values at different points in the circuit
  - This is because the current **splits** at a junction
  - Therefore, the electrons have different paths they can take
- The **sum** of the current in the **individual** branches is equal to the **total** current before (and after) the branches



Current splits at a junction into individual branches

Why is current conserved at a junction in a circuit?

- At a junction, the current is always **conserved**
  - This means the amount of current flowing into the junction is equal to the amount of current flowing out of it
  - This is because the **charge** is conserved
- Current does not always split equally – often there will be more current in some branches than in others
  - The current in each branch will only be identical if the resistance of the components along each branch is **identical**
- Current behaves in this way because it is the **flow of electrons**:
  - Electrons, or any charge, cannot be created or destroyed

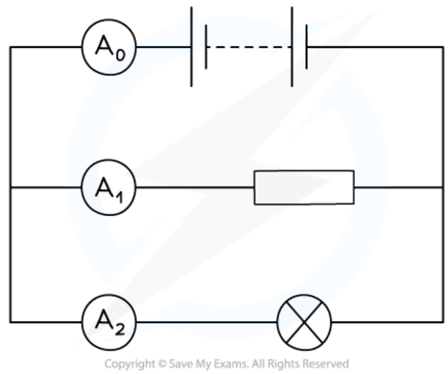
- This means the total number of electrons (and hence current) going around a circuit must remain the same
- When the electrons reach a junction, however, some of them will go one way and the rest will go the other

Your notes

Your notes

Worked example

In the circuit below, ammeter A_0 shows a reading of 10 A, and ammeter A_1 shows a reading of 6 A.



What is the reading on ammeter A_2?

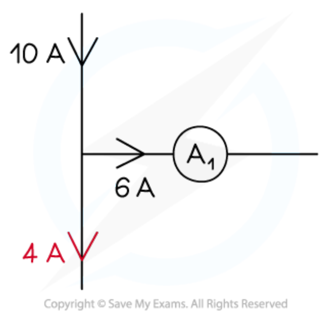
Answer:

Step 1: Recall what happens to the current at a junction

- At a junction, the current splits, but is always conserved
- This means that the total amount of current flowing into a junction is equal to the total amount flowing out

Step 2: Consider the first junction in the circuit where the current splits

- The diagram below shows the first junction in the circuit



Step 3: Calculate the missing amount of current

- Since 10 A flows into the junction (the total current from the battery), 10 A must flow out of the junction
- The question says that 6 A flows through ammeter A_1 so the remaining current flowing through ammeter A_2 must be:

$$10\text{ A} - 6\text{ A} = \mathbf{4\text{ A}}$$

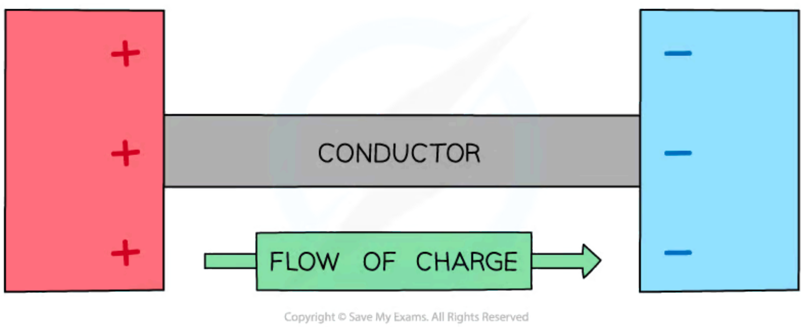
- Therefore, **4 A** flows through ammeter A_2

Your notes

Examiner Tip

The direction of current flow is super important when considering junctions in a circuit.

You should remember that current flows from the **positive** terminal to the **negative** terminal of a cell / battery. This will help determine the direction current is flowing 'in' to a junction and which way the current then flows 'out'.



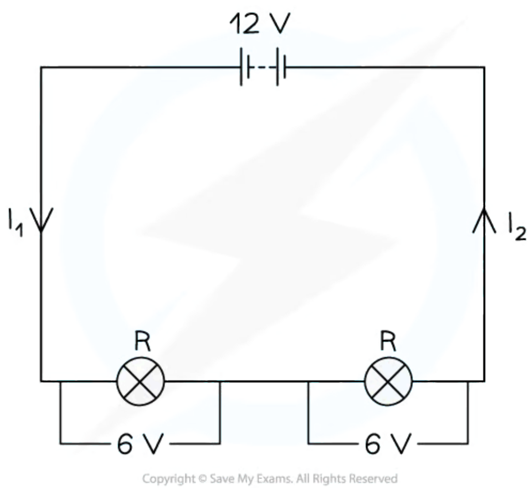
Voltage in Series & Parallel

Your notes

Voltage in series & parallel

Voltage in series

- In a series circuit, the **total voltage** of a power supply is **shared** between the components



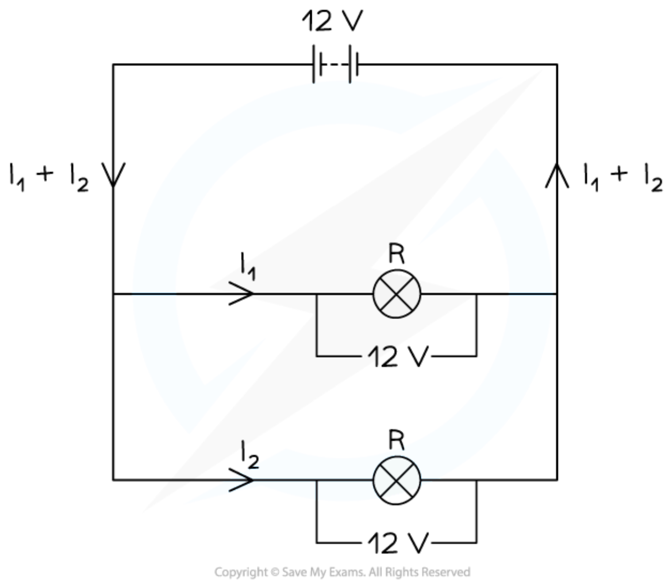
Lamps connected in a series circuit share the potential difference from the battery

- For two identical components (with equal resistance), the voltage across them will be:
  - the **same**
  - equal to **half the total voltage** of the power supply
- For two non-identical components (with different values of resistance), the voltage will be:
  - **higher** across the component with the higher resistance
  - **lower** across the component with lower resistance

Voltage in parallel

- In a parallel circuit, the **total voltage** across each branch is the **same** as the voltage of the power supply

Your notes



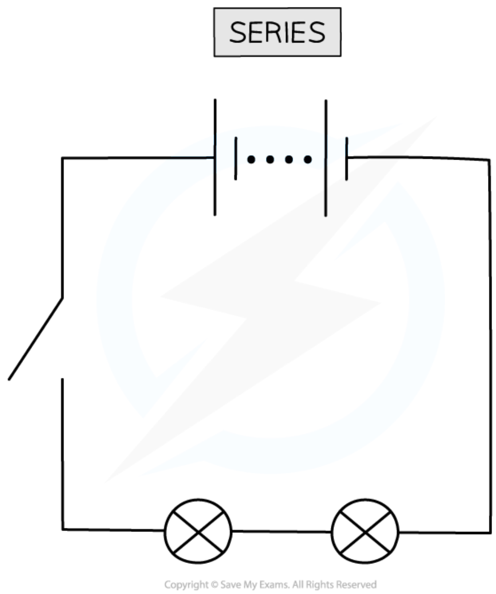
Lamps connected in a parallel circuit all have the same voltage across them

Your notes

Advantages & disadvantages

Advantages and disadvantages of a series circuit

- A **series** circuit consists of a string of two or more components connected in a loop



In a series circuit, only one switch is needed to control all of the lamps. This can be seen as an advantage or as a disadvantage

Advantages of a series circuit

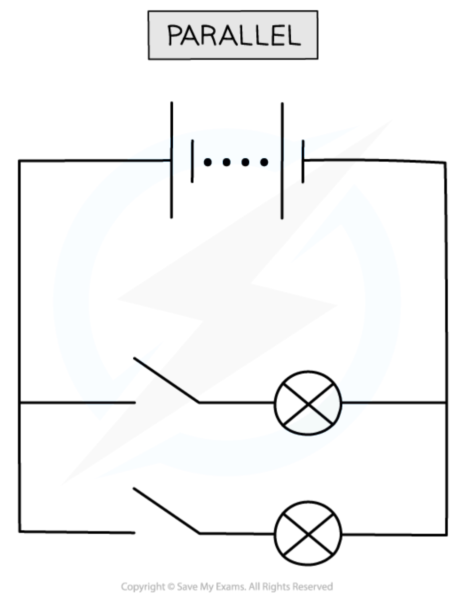
- All of the components are controlled by a **single switch**
- Fewer wires are required

Disadvantages of a series circuit

- The components cannot be controlled **separately**
- If one component breaks, all other components stop working

Advantages and disadvantages of parallel circuits

- A **parallel** circuit consists of two or more components attached across **different branches** of the circuit



In a parallel circuit, the lamps are connected in parallel and can be switched on and off by their own switches

Advantages of a parallel circuit

- The components can be **individually controlled** using their own switches
- If one component breaks, then the others will continue to function

Disadvantages of a parallel circuit

- Many more wires are involved which can be more complicated to set up
- All branches have the same voltage as the supply making it more difficult to control the voltage across individual components

Examiner Tip

You may have noticed that for a parallel circuit, all of the components can be controlled by a single switch - like a series circuit. Nevertheless, the exam board still considers this an advantage of **series** circuits.

Note that the current does not always split equally in a parallel circuit – often there will be more current in some branches than in others. The current in each branch will only be identical if the **resistance** of the components along each branch are identical. However, the voltage across two components connected in parallel is always the **same**

Your notes

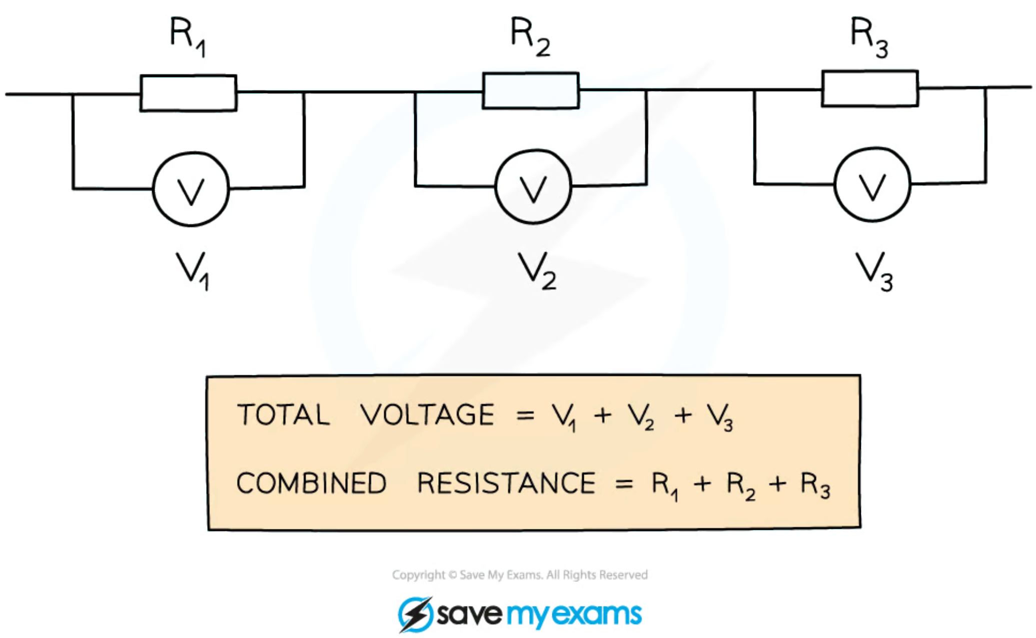
Resistors in Series

Resistors in series

- When two or more resistors are connected in series, the **total resistance** is equal to the **sum** of their individual resistances
- For two resistors of resistance R_1 and R_2, the total resistance can be calculated using the equation:

$$R = R_1 + R_2$$

- Where:
  - R is the total resistance, in ohms (Ω)
- Increasing the number of resistors **increases** the overall resistance
  - The charge now has **more** resistors to pass through
- The **total voltage** is also the **sum** of the voltages across each of the **individual resistors**



Three resistors connected in series. The total voltage is the sum of the individual voltages, and the total resistance is the sum of the three individual resistances

Summary of series and parallel circuits

- For components connected in **series**:
  - the **current** is the same at all points and in each component
  - the **voltage** of the power supply is shared between the components
  - the **total resistance** is the sum of the resistances of each component
- For components connected in **parallel**:

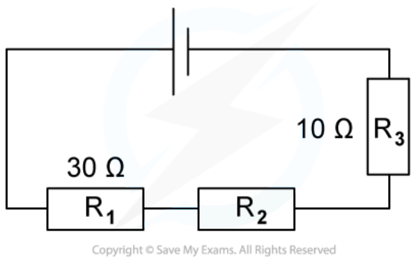
- the **current** from the supply splits in the branches
- the **voltage** across each branch is the same
- the **total resistance** is less than that of each component

Your notes

Worked example

The combined resistance *R* in the following series circuit is 60 Ω.

What is the resistance value of R_2?



A 100 Ω **B** 30 Ω **C** 20 Ω **D** 40 Ω

ANSWER: C

Step 1: Write down the equation for the combined resistance in series

$$R = R_1 + R_2 + R_3$$

Step 2: Substitute the values for total resistance *R* and the other resistors

$$60\ \Omega = 30\ \Omega + R_2 + 10\ \Omega$$

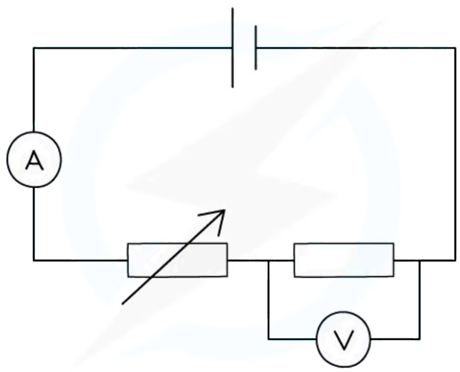
Step 3: Rearrange for R_2

$$R_2 = 60\ \Omega - 30\ \Omega - 10\ \Omega = \mathbf{20\ \Omega}$$

Your notes

Worked example

Dennis sets up a series circuit as shown below.



The cell supplies a current of 2 A to the circuit, and the fixed resistor has a resistance of 4 Ω.

(a) How much current flows through the fixed resistor?

(b) What is the reading on the voltmeter?

Answer:

Part (a)

Step 1: Recall that current is conserved in a series circuit

- Since current is conserved in a series circuit, it is the **same size** if measured anywhere in the series loop
- This means that since the cell supplies 2 A to the circuit, the current is 2 A everywhere
- Therefore, **2 A** flows through the fixed resistor

Part (b)

Step 1: List the known quantities

- Current, I = 2 A
- Resistance, R = 4 Ω

Step 2: State the equation linking potential difference, resistance and current

- The equation linking potential difference, resistance and current is:

$$V = I \times R$$

Step 3: Substitute the known values into the equation and calculate the potential difference

$$V = 2 \times 4 = 8 \text{ V}$$

- Therefore, the voltmeter reads **8 V** across the fixed resistor

Your notes

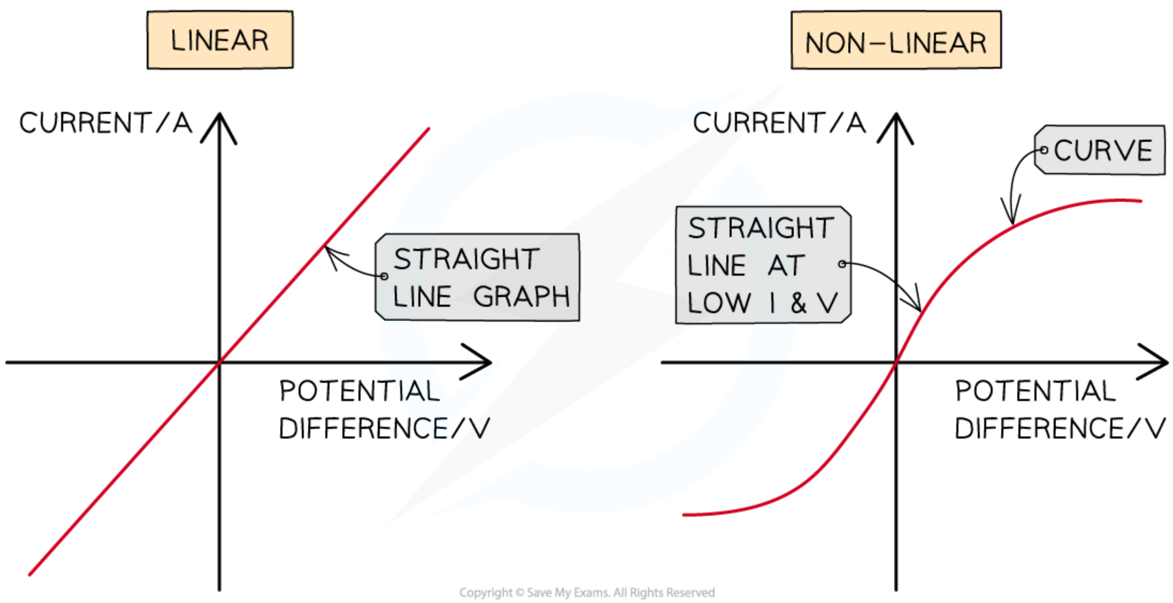
IV Graphs

Your notes

IV Graphs

- When the voltage *V* across a component is varied, the current *I* flowing through it may vary **linearly** or **non-linearly**
- The relationship between current and voltage of a component can be shown on an *IV* graph
- When the relationship between current and voltage is **linear**:
  - the *IV* graph is a **straight line** which passes through the **origin**
  - the resistance is **constant**
- When the relationship between current and voltage is **non-linear**:
  - the *IV* graph that is **not** a straight line
  - the resistance is **not** constant

Linear and non-linear *IV* graphs

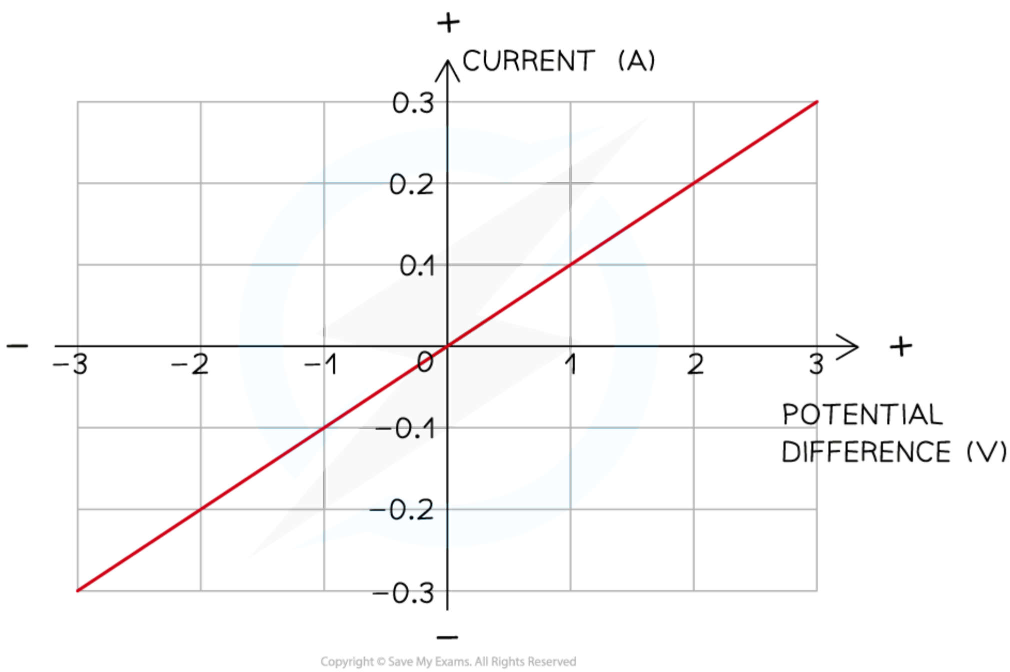


Linear IV graphs are straight lines through the origin, indicating a constant resistance. Non-linear IV graphs are curved, indicating a variable resistance

- Components with **linear** *IV* graphs include:
  - fixed resistors (at constant temperature)
  - wires (at constant temperature)
- Components with **non-linear** *IV* graphs include:
  - filament lamps
  - diodes
  - LDRs
  - thermistors

IV graph for a wire or a resistor

- The relationship between current and voltage for a wire or fixed resistor is linear, or **directly proportional**, which means
  - the IV graph is a straight line, so voltage and current increase (or decrease) by the **same** amount
  - the slope of the graph is constant, so resistance is **constant**



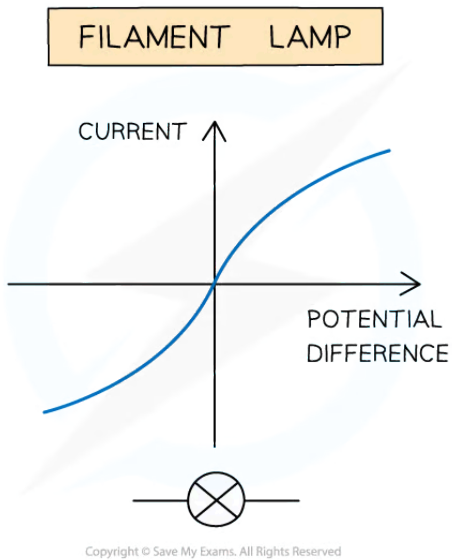
The current is directly proportional to the potential difference (voltage) as the graph is a straight line through the origin

IV graph for a filament bulb

- The relationship between current and voltage for a filament lamp is non-linear, or **not** directly proportional, which means
  - the *IV* graph is not a straight line, so voltage and current do **not** increase (or decrease) by the same amount
  - the slope of the graph is not constant, so resistance **changes**
- The *IV* graph for a filament lamp shows as voltage increases
  - the current increases at a proportionally **slower** rate
  - the resistance **increases**; the flatter the slope, the higher the resistance

Your notes

Your notes

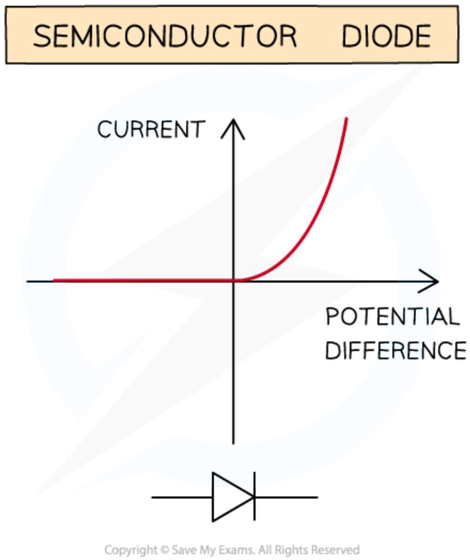


IV graph for a filament lamp

- As current through a filament lamp increases, the resistance **increases** because:
  - the higher current causes the **temperature** of the filament to increase
  - the higher temperature causes the atoms in the metal lattice of the filament to **vibrate** more
  - this causes an increase in resistance as it becomes more difficult for **free electrons** (the current) to pass through
  - since resistance **opposes** the current, this causes it to increase at a **slower** rate

IV graph for a diode

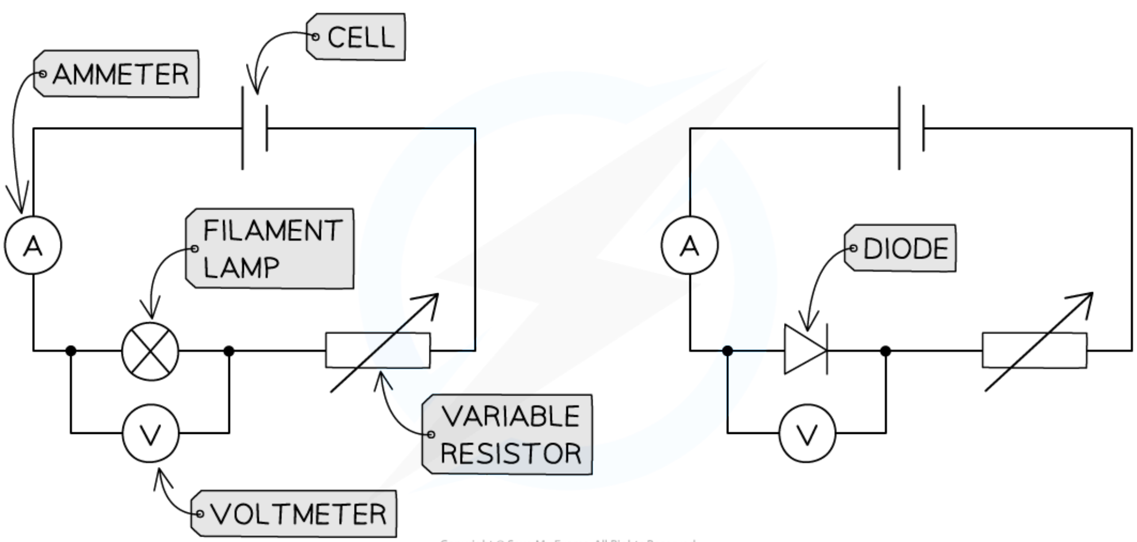
- A diode allows current to flow in **one** direction only
  - This is called **forward bias**
- In the reverse direction, the diode has very **high resistance**, and therefore **no** current flows
  - This is called **reverse bias**
- When the current is in the direction of the arrowhead symbol, this is **forward bias**
  - On the IV graph, this is shown by a sharp increase in voltage and current on the right side of the graph
  - This shows the resistance is very **low**
- When the diode is switched around, this is **reverse bias**
  - On the IV graph, this is shown by a zero reading of current or voltage on the left side of the graph
  - This shows the resistance is very **high**



IV graph for a semiconductor diode

Investigating the relationship between current and voltage

- In order to investigate the relationship between **current** and **voltage** different components, the following equipment is required:
  - an **ammeter** - to measure the current through the component
  - a **voltmeter** - to measure the voltage across the component
  - a **variable resistor** - to vary the current through the circuit
  - a **power source** - to provide a source of potential difference (voltage)
  - **wires** - to connect the components together in a circuit
- The image below shows the circuits set up to obtain *IV* graphs for a filament lamp and a diode



These circuits enable the investigation of current and voltage for a filament lamp or diode to be investigated

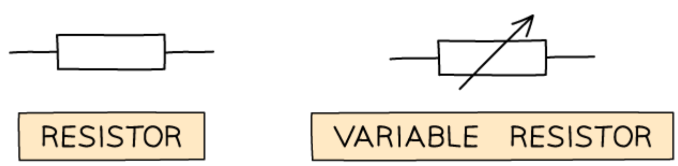
- The **current** is the **independent variable**
  - The **variable resistor** is used to change the current flowing through the filament lamp / diode
- The **voltage** is the **dependent variable**
  - The **voltmeter** is used to measure the voltage across the filament lamp / diode
- Recording measurements of current and voltage as the current increases enables an *IV* graph to be plotted for each component

Resistance

- Resistance is the **opposition** to the flow of **current**
  - The **higher** the resistance of a circuit the **lower** the current

- Resistors come in two types:
  - **Fixed** resistors
  - **Variable** resistors

- Fixed resistors have a resistance that remains **constant**
- Variable resistors can **change** the resistance by changing the **length** of wire that makes up the circuit
  - A **longer** length of wire has **more resistance** than a shorter length of wire



Fixed and variable resistor circuit symbols

Your notes

Electrical Components

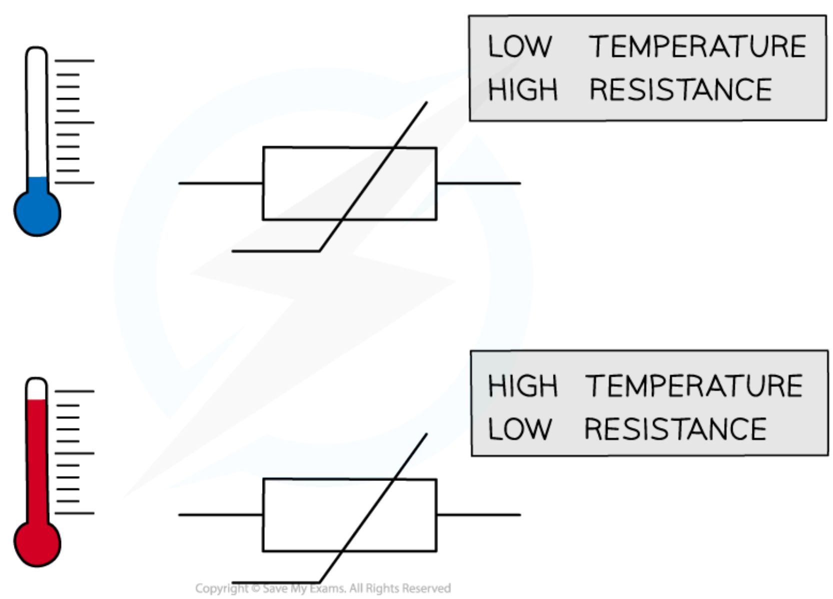
Your notes

Thermistors & LDRs

- Environmental conditions, such as temperature and light intensity, can influence the resistance of resistors, such as
  - Thermistors
  - Light-dependent resistors (LDRs)

Thermistors

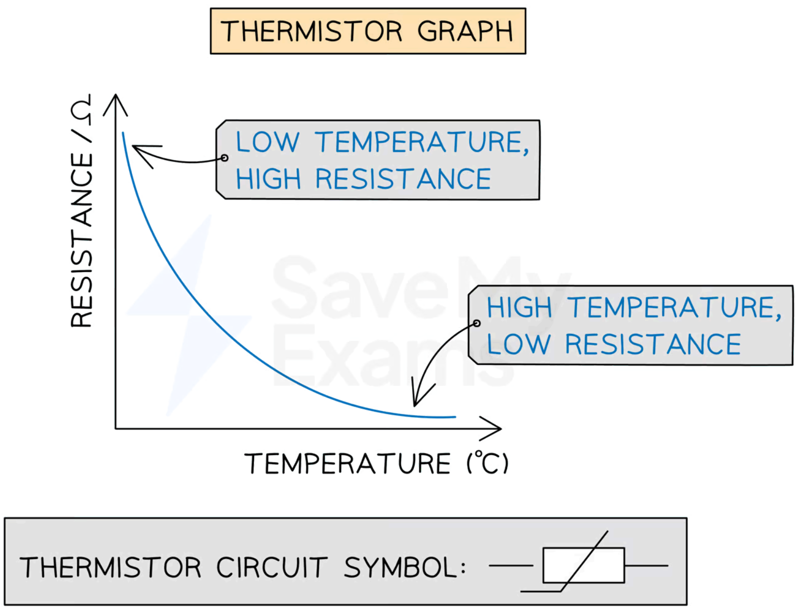
- The resistance of a **thermistor** depends on its **temperature**
- The resistance of a thermistor is **high** in **cold** conditions and **low** in **hot** conditions
  - As the temperature **increases** the resistance of a thermistor **decreases**
  - As the temperature **decreases** the resistance of a thermistor **increases**



The resistance of a thermistor depends on its temperature

- The relationship between resistance and temperature for a thermistor can be shown on a graph

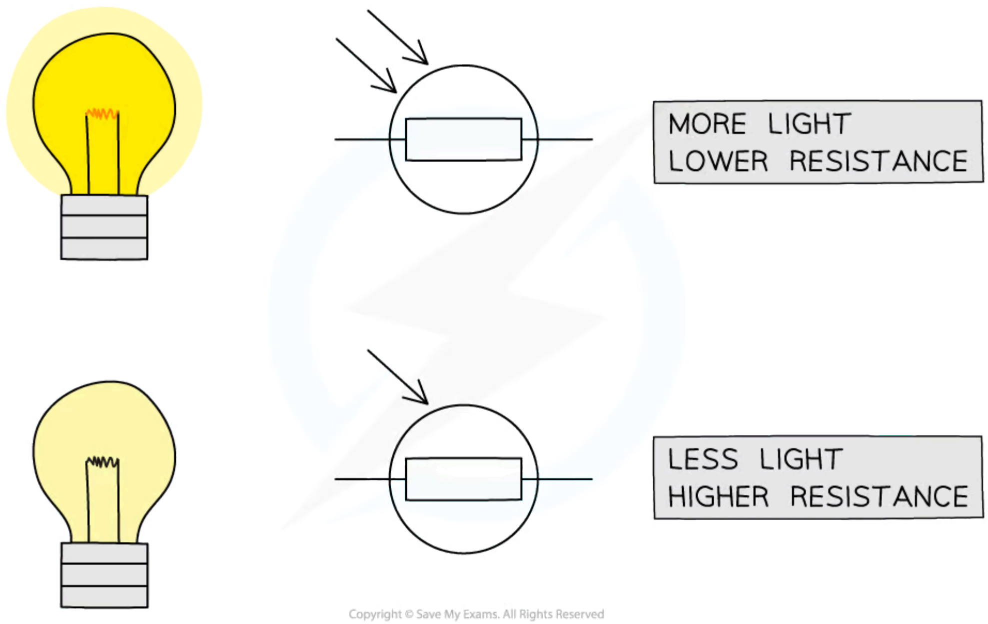
Your notes



The graph of resistance against temperature for a thermistor shows a curve indicating these quantities are inversely proportional to each other

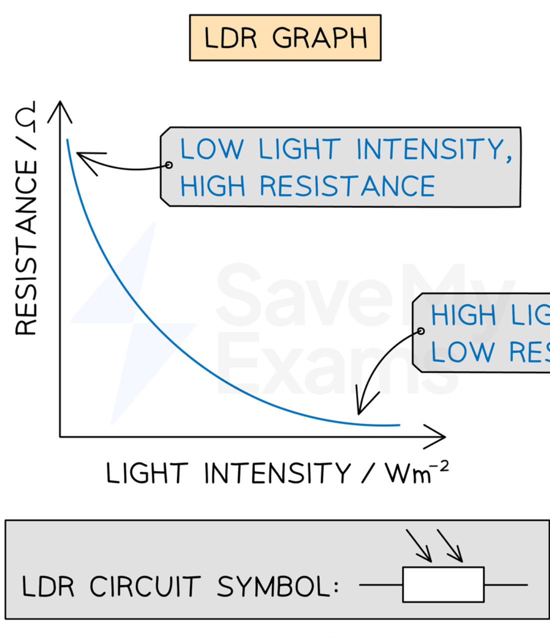
Light-dependent resistors (LDRs)

- The resistance of a **light-dependent resistor** (LDR) depends on the **light intensity** on it
- The resistance of an LDR is **high** in **dark** conditions and **low** in **bright** conditions
  - As the light intensity **increases** the resistance of an LDR **decreases**
  - As the light intensity **decreases** the resistance of an LDR **increases**



The resistance of an LDR depends on the intensity of light on it

- The relationship between resistance and light intensity for an LDR can be shown on a graph



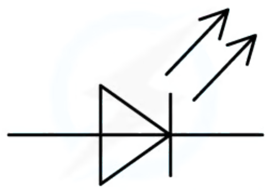
The graph of light intensity against temperature for an LDR shows a curve indicating these quantities are inversely proportional to each other

Lamps & LEDs

- Lamps and light-emitting diodes (LEDs) **illuminate** (light up) when a **current** flows through them
- This makes them useful for indicating the presence of a current in a circuit

Light-emitting diodes (LEDs)

- LEDs are a type of **diode**
  - This means they only allow current to flow through them in one direction
  - Therefore, in a circuit, an LED will only light up if it is placed in the correct direction
- The circuit symbol for an LED is as follows:



LEDs can be used to indicate the presence of a current as they illuminate when current flows through them. The same is true for lamps

Examiner Tip

Make sure you learn the various symbols mentioned on this page. Many of them are very similar with small differences denoting what they do:

- Two arrows pointing towards a symbol mean that it is **light-dependent**
- Two arrows pointing away mean that it is **light-emitting**

Symbols are sometimes drawn with circles around them (e.g. the LDR). These circles are often optional (although not in the case of meters and bulbs).

Your notes